

Jeevan Sanchez

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EDUCATION

Queen's University

Bachelor of Applied Science in Computer Engineering

- Direct Entry Computer Engineering

Exp. May 2029

Kingston, ON

EXPERIENCE

Queen's Rocket Engineering Team

Propulsion Integration Systems Designer

Avionics Lead (since September 2026)

September 2025 - Present

Kingston, ON

- Co-leading the avionics subteam, coordinating **firmware and embedded systems development** for **propulsion control, sensing, and communications**.
- Developed **embedded C/C++ firmware** and **hardware drivers** for ADCs, Hall-effect sensors, and real-time actuator control across **analog, SPI, I²C, and CAN/TWAI** interfaces.
- Designed and integrated **firmware architectures** for **STM32/ESP32-based systems** using **STM32CubeMX, STM32 HAL, and PlatformIO**, supporting modular sensor, actuator, and communications interfaces.
- Supported **firmware bring-up and system integration**, troubleshooting **hardware, firmware, peripheral interfaces, and communications** issues through **functional testing** and iterative debugging with the team.

Queen's Engineering Society IT-OPS Team

IT-Operations Team Member

September 2025 - May 2026

Kingston, ON

- **Provide first-level technical support** within a small team maintaining Engineering Society systems.
- **Researched an ML-powered RAG pipeline** using **LangChain and OpenAI** to automate IT support workflows and improve **knowledge retrieval and ticket resolution**.

PROJECTS

Penguin ([Link](#))

Intelligent Robotics | ESP32-S3, C++, Arduino, PlatformIO, KiCad, OpenCV, Next.js)

May 2026 - Present

Toronto, ON

- **Engineered end-to-end embedded firmware architecture** and custom PCB for a **hybrid wheel-legged rover**, executing **full HW/SW bring-up and system integration**.
- **Developed modular C++ firmware** in **PlatformIO**; authored **low-level peripheral drivers (I2C, PWM)** and type-safe **command/state serialization structures** for motors, IMU, ToF units, and other peripherals.
- **Architected a deterministic real-time control loop and finite state machine (FSM)** driving multi-axis closed-loop **PID locomotion** and sensor fusion.
- **Integrating an OV2640 camera (DVP/DMA)** with an **OpenCV perception pipeline** alongside an asynchronous Wi-Fi **WebSocket telemetry** and command stack.

Vireo ([Link](#))

September 2026 - Present

Audio Interface, Signal Processing | (KiCad, STM32, C/C++, Analog/Digital Electronics)

Toronto, ON

- **Designing** a multi-channel instrument audio interface integrating precision multi-channel **ADCs**, analog signal conditioning circuitry, and an **STM32** embedded platform.
- **Authoring** modular C/C++ firmware using **STM32CubeMX** and **STM32 HAL** for real-time audio acquisition, on-device **DSP**, and class-compliant **MIDI** communication.
- **Accelerating** zero-copy **DMA**-driven high-speed **ADC** transfers and double-buffered processing pipelines to ensure low-latency, deterministic execution.

HomeCtrl ([Link](#))

August 2025 - July 2026

Smart Home-Automation Controller | (ESP32, KiCad, WebSocket, Arduino C/C++, Next.js)

Toronto, ON

- **Architected** a full-stack **home automation platform**, designing custom **ESP32-S3** hardware, **embedded firmware**, and system architecture.
- **Programmed concurrent C++ firmware** in **FreeRTOS**, implementing **multi-tasking peripheral drivers**, **FSMs**, and **deterministic control loops**.
- **Engineered bi-directional real-time telemetry** and control over **Wi-Fi WebSockets** linking the ESP32 to a **Next.js** dashboard with **Alexa** voice integration.
- **Validated** end-to-end system reliability through **iterative hardware bring-up**, **signal debugging**, and **comprehensive functional testing**.

CERTIFICATIONS

Microsoft Azure AI Fundamentals

Microsoft | Azure, AI/ML | AI-900

August 2025

Toronto, ON

CS50's Introduction to Artificial Intelligence with Python

Harvard University | AI/ML Theory, Python, sklearn | CS50AI

July 2025

Toronto, ON

CS50's Introduction to Computer Science

Harvard University | C Programming, Python, JavaScript, SQL | CS50

July 2024

Toronto, ON

AWARDS

First Place at UofT UTRA Hacks Hackathon

Health System with Integrated Robotics and Firmware | (Arduino C/C++, Next.js, AI/ML)
University of Toronto

February 2025

Toronto, ON

First Place at UofT SEEK JR. Hackathon

Autonomous Martian Rover | (Arduino C/C++)
University of Toronto

February 2024

Toronto, ON